Project Development Phase

Preprocessing Steps & Business Questions

Field	Details
Team Lead	Konkimalla Bhagyasri Lasya
Team Members	Bhavitha Medakayala, Yogiswari Sanapala, Divya kauluri, Nagaram praveenya
Domain	India's Agriculture Crop Production Analysis
Tools Used	Tableau
Dataset	Historical Indian Crop Production Data, 1997–2021
Published Dashboard	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject

Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and what to invest in the agricultural market. The Investor's View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

1. Data Preprocessing Steps

Before any chart was built, the raw government crop production extract (345,407 rows, 1997–2021) was cleaned and enriched using Python 3.12 with the Pandas library.

Step	Action	Why It Was Needed
1	Load Raw Extract	Single flat file with no relational structure; every subsequent check has to be built manually
2	Remove Duplicates & Blanks	Prevents double-counting or silently broken aggregations like SUM(Production)
3	Standardise Names	Prevents inconsistent state spelling from splitting one state into multiple map regions
4	Derive Crop_Year_Numeric	A numeric year is required to drive a continuous time axis on trend charts
5	Compute Yield	Yield does not exist in the raw extract but is essential for measuring productivity
6	Compute Production_Normalized	Production_Units differ by crop (Tonnes, Bales, Nuts); normalisation avoids conflating incompatible units
7	Convert to Millions	Raw totals run into the hundreds of millions; scaling keeps chart axes and KPI cards readable
8	Export Cleaned CSV	skillwallet_crop_cleaned.csv is the single file Tableau connects to

Data Quality Checks Applied

- Row-count reconciliation: cleaned file's row count compared against raw file's row count after removing duplicates/blanks
- Range checks: Area and Production values checked for negative or zero entries indicating a data entry error
- Unit consistency: Production_Units cross-checked against Crop (e.g., Coconut consistently in Nuts, Cotton in Bales)
- Partial-year flag: 2021 records flagged as a partial-year data limitation, since year-end reporting was incomplete

2. Business Questions & Visualizations

Q#	Business Question (Persona)	Answered By	Key Insight
Q1	National Overview — what is India's overall agricultural performance at a glance?	KPI Summary panel	Avg. Yield 8,487.41; Area 4,031.13 M ha; Production 330,997.84 M (1997–2021)

Q2	Trend — has national crop yield improved or declined over 25 years? (Policy Analyst)	Trend of Agricultural Yield line chart	Yield rises from ~4.0 (1997) to a peak above 4.75 (2017–2019); final-year dip is a 2021 reporting artifact

Q#	Business Question (Persona)	Answered By	Key Insight
Q3	Which states/regions produce the most, and where should investment focus? (Policy Analyst)	State-wise Production Map	Shades every state by total production, making regional disparities visible at a glance
Q4	Which crops occupy the most cultivated land nationally? (Farmer)	Cultivated Area by Crop bar chart	Rice and Wheat dominate national land use, followed by Cotton(lint) and Soyabean
Q5	Which states have the largest agricultural land base? (Investor)	State-wise Cultivated Area Comparison	Uttar Pradesh, Madhya Pradesh, Rajasthan, and Maharashtra lead by a wide margin
Q6	Which states generate the highest production volume? (Investor)	Leading States by Agricultural Production bar chart	Read alongside Q8 to correctly interpret the unit basis behind each state's ranking
Q7	How has total national production grown year over year? (Policy Analyst)	Year-wise Total Agricultural Production bar chart	Steady growth through the early 2010s, followed by fluctuation rather than continued acceleration
Q8	Are production rankings distorted by mixing incompatible measurement units? (Investor/Analyst)	Production Distribution by Unit Type donut chart	Majority of raw production figures come from crops measured in Nuts (Coconut) — explains why Kerala ranks so highly in Q6
Q9	Which growing season should I prioritise for the crop I want to plant? (Farmer)	Season-wise Production pie chart	Whole Year crops dominate (317,910.43 M), followed by Rabi (4,264.02 M) and Winter (598.30 M)